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Transcutaneous electrical acupoint stimulation applied in lower limbs decreases the incidence of paralytic ileus after colorectal surgery: A multicenter randomized controlled trial

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ABSTRACT

Background: Postoperative paralytic ileus prolongs hospitalization duration, increases medical expenses, and is even associated with postoperative mortality; however, effective prevention of postoperative paralytic ileus is not yet available. This trial aimed to assess the preventative effectiveness of transcutaneous electrical acupoint stimulation applied in the lower limbs on postoperative paralytic ileus incidence after colorectal surgery.

Methods: After ethics approval and written informed consent, 610 patients from 10 hospitals who were scheduled for colorectal surgery between May 2018 and September 2019 were enrolled. Patients were randomly allocated into the transcutaneous electrical acupoint stimulation (stimulated on bilateral Zusanli, Shangjuxu, and Sanyinjiao acupoints in lower limbs for 30 minutes each time, total 4 times) or sham (without currents delivered) group with 1:1 ratio. The primary outcome was postoperative paralytic ileus incidence, defined as no flatus for >72 hours after surgery.

Results: Compared to the sham treatment, transcutaneous electrical acupoint stimulation lowered the postoperative paralytic ileus incidence by 8.7% (32.3% vs 41.0%, $P = .026$) and decreased the risk of postoperative paralytic ileus by 32% (OR, 0.68; $P = .029$). Transcutaneous electrical acupoint stimulation also shortened the recovery time to flatus, defecation, normal diet, and bowel sounds. Transcutaneous electrical acupoint stimulation treatment significantly increased median serum acetylcholine by 55% ($P = .007$) and interleukin-10 by 88% ($P < .001$), but decreased interleukin-6 by 47% ($P < .001$) and inducible nitric oxide synthase by 42% ($P = .002$) at 72 hours postoperatively.

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Conclusion: Transcutaneous electrical acupoint stimulation attenuated the postoperative paralytic ileus incidence and enhanced gastrointestinal functional recovery, which may be associated with increasing parasympathetic nerve tone and its anti-inflammatory actions.

Introduction

Colorectal cancer is one of the most common cancers in China, and the 5-year survival of patients with early operative treatment is more than 90%.[1,2] Postoperative paralytic ileus (POI) is a common postoperative complication after colorectal surgery, which increases hospital stay length by 25%, hospital cost by 15%, and postoperative 30-day complication-associated mortality and also affects long-term quality of life.[3] POI is characterized by nonmechanical temporary gastrointestinal dysfunction with various symptoms, including delayed flatus and defecation, abdominal pain, bloating, nausea, vomiting, and intolerance to an oral diet. The reported incidence of POI is varied, with some reports indicating a considerably high incidence.[4,5] Lack of flatus for >72 hours after surgery has been suggested to be POI, although other gastrointestinal symptoms including loss of appetite have also been considered to be POI.[6]

The noxious stimulation of intestinal manipulation during surgery activates prevertebral adrenergic neurons to release noradrenaline and suppresses parasympathetic nerve tone.[7] Surgical trauma also induces an inflammatory response via the activated resident macrophages release of inflammatory cytokines and chemokines.[8] The latter attract leukocytes to invade the intestinal muscularis externa and induce time-dependent induction of cytokine release to impair the contractile activity of the smooth muscle cell,[9] resulting in the development of POI. Symptomatic treatment is the main clinical management of POI; effective interventions are not yet available. For example, first-line symptomatic medications (brain peptides, gastrografin, and cholinergic drugs) and powerful defecation-inducing drugs (enemas and laxatives) are not beneficial for gastrointestinal functional recovery due to their side effects, such as cardiovascular adverse reactions, immunosuppression, gastrointestinal tract irritation, and even electrolyte disorders.[10,11] However, activation or mimicry of the cholinergic anti-inflammatory pathway by stimulation of the vagal nerve was reported as a potential treatment to inhibit the activation of resident macrophages and restore intestinal function.[8]

Acupuncture, a safe and effective nonpharmacotherapy, can directly or indirectly increase the efferent vagal nerve activity and even deactivate inflammatory macrophages.[12] Acupuncture has been demonstrated to improve nonsurgical gastrointestinal disorders such as chronic severe functional constipation,[13] functional dyspepsia,[14] and inflammatory bowel disease,[15] thereby showing a great potential to prevent or treat POI.[16] Transcutaneous acupoint electrical stimulation (TEAS) is the combination of transcutaneous electrical nerve stimulation and acupoint therapy.[17,18] TEAS produces a perceptible sensation through electrodes placed on skin acupoints. Its electrode slice has a single conductive emboss and 2 connecting wires, thereby enabling noninvasive, accurate, and convenient stimuli to be delivered. Compared to conventional acupuncture or electroacupuncture, TEAS has no risk of infection, contamination, patients' psychological dread, or operator bias.[18] The effectiveness of TEAS in preventing POI reported in previous studies was largely inconclusive due to unclear POI definition, small sample size without proper power calculation (the largest trial to date included only 110 patients), inadequate bias control, and unclear underlying mechanism.[19,20] Therefore, we conducted this large-sample-size, multicenter, randomized sham-controlled study, which was in accordance with clinical research Consohdated Standards of Reporting Trials (CONSORT) and the STandards for Reporting Interventions in Clinical Trials of Acupuncture (STRICTA) acupuncture clinical trial. The primary outcome was the incidence of POI, defined as no flatus for >72 hours after surgery as reported by the European Association of Colorectal Surgeons.[21] To date, our study is the largest sample size trial to explore the effectiveness of TEAS on preventing POI and the first to investigate its underlying mechanisms.

Methods

Study design and participants

This multicenter, parallel-group, and randomized controlled trial complied with the Declaration of Helsinki, was approved by the Ethics Committee of the First Affiliated Hospital of Xi'an Jiaotong University (XJTU1AF2016LSL-035), and was registered at ClinicalTrials.gov (NCT03086304). Patients scheduled for colorectal cancer surgery at 10 hospitals in China from May 2018 to September 2019 were screened for potential enrollment after written informed consent was obtained. Inclusion criteria were: (1) surgery under general anesthesia; (2) age ≥18 years and American Society of Anesthesiologists I—III; (3) patients' ability to understand and sign the written informed consent form; and (4) not involved in other clinical studies. Exclusion criteria were: (1) history of intestinal surgery; (2) skin damage or infection at any acupoints, allergic to adhesive tape; (3) intraoperative enterostomy; (4) pregnant or breast-feeding; and (5) patients' refusal to participate in the study. All investigators were trained according to the standardized acupoints selection and TEAS manipulation by a senior acupuncturist with 20 years of practicing experience. The detailed trial protocol is available online as Appendix 1. In an effort to improve trial compliance, the potential benefits of participating in the study, including noninvasive treatment, no additional costs, and POI prevention, were explained to the patients and their caregivers/family members. To obtain accurate first gastrointestinal function recovery time, the patients and their caregivers/family members were required to agree to accurately record data.

Anesthesia, surgery, and patients' randomization

Patients were premedicated before anesthesia induction with antiemetics (palonosetron 0.075 mg and dexamethasone 5 mg), salivary suppressor (penehyclidine hydrochloride 0.5 mg), and nonsteroidal anti-inflammatory drugs (flurbiprofen axeril 50 mg). After the transverse abdominis plane block (0.375% ropivacaine 20 mL) was carried out, patients were inducted with etomidate (0.2 mg/kg), sufentanil (0.3 μg/kg), and rocuronium bromide (1 mg/kg) and intubated. Etomidate was used for induction instead of propofol because of a putatively favorable hemodynamic profile.[22] The patients' lungs were mechanically ventilated. Remifentanil (0.1 μg/kg/h), cisphenyl sulfatracurium (0.1 mg/kg/h), dexmedetomidine (initial dose 1 μg/kg for 10 min, maintenance dose 0.4 μg/kg/h), and sevoflurane and/or propofol, as necessary, were used for anesthesia maintenance to keep bispectral index 40–60 throughout the surgery. Surgical decisions such as colectomy or proctectomy, laparoscopy or laparotomy, postoperative diet, and treatment were made by the surgeons as routine clinical practice. Patients' eligibility was assessed by 2-step screenings for final enrollment and randomization. The initial screening and recruitment were undertaken 1 day before surgery, and the final enrollment and randomization were conducted at the completion of surgery and general anesthesia. SPSS software was used for block randomization in a 1:1 ratio with block size of 4. This random method will ensure that the number of patients in the 2 groups will be essentially equal with highly efficient enrollment. After rescreening for eligibility at the completion of surgery, the central randomization administrator opened the sealed envelope and made the group allocation to the intervention executor in the corresponding center. All participants and study personnel, including researchers who collected data and anesthesiologists and surgeons who cared for patients, were blinded to the patient's group allocation.

Interventions

TEAS was administrated a total of 4 times in each patient: the first time was in the postoperative recovery unit (PACU) when patients reached a Steward recovery score of $\geq$4 and were able to cooperate with the treatment, and the remaining 3 times were once each day between 9 AM and 11 AM on the first to third postoperative days in the ward. The Steward score (assessing conscious level, airway health, and muscle strength) was used to assess the patient's recovery from anesthesia. The Zusanli (ST36), Shangjuxu (ST37), and Sanyinjiao (SP6) acupoints at both legs (Supplementary Fig S1 in Appendix 2) were identified to deliver TEAS. Electrode pads were then affixed to these acupoints. Once the effective tingling sensation "de qi," which indicates ample current at a frequency of 2 to 10 Hz stimulation (TEAS stimulator, SDZ-V; Hwato, China), was achieved at these acupoints, TEAS was adjusted to the maximum intensity tolerated by the patient. This was then maintained for 30 minutes. To minimize bias, patients were blinded to study group allocation. Patients in the sham group received identical settings and manipulation, as did those in the TEAS group, except that the wire between the electrode pads and TEAS stimulator was severed to prevent current delivery.

Outcomes

The primary outcome of the study was the incidence of POI. No flatus for >72 hours after surgery was diagnosed as POI. The secondary outcomes were various POI symptoms, including the time to first postoperative flatus, defecation, bowel sound recovery, normal diet recovery, and the incidence of abdominal pain, distention, nausea, and vomiting, all of which were recorded from the first to third postoperative days and 30-day telephone follow-up. The interval time was from when the Steward recovery score reached 4 points immediately after surgery to the first occurrence time indicated above on the first postoperative day.

Blood sample analysis

After patients in both groups received 4 interventions, the blood samples were collected in serum tubes at the postoperative 72 hours, immediately placed on ice, centrifuged at 4°C, and stored at −80°C. The blood samples were randomly selected from both groups with a rate of 10% of the study participants. This sampling rate based on power calculation was done from the preliminary data obtained from the feasibility study. Serum acetylcholine (Ach) and interleukin-10 (IL-10) were measured with enzyme-linked immunoassay kits (Shanghai Lianmai Bioengineering Co. Ltd, Shanghai, China), and serum interleukin-6 (IL-6) and inducible nitric oxide synthase (iNOS) were also determined with enzyme-linked immunoassay kits (Cusabio Biotech Co. Ltd, Wuhan, China). These concentrations were compared between the TEAS and sham groups, between the POI and non-POI groups, and between laparoscopic (closed) and proctectomy (open) surgery.

Statistical analysis

Power calculation: The POI incidence in a recent study was reportedly approximately 48%,[23] and yet the POI incidence under the intervention of acupuncture has been rarely reported.[24] Therefore, the pilot feasibility study was carried out and the POI incidence was reduced by 12% with the TEAS treatment calculated from the preliminary data. With a significance set at .05 and power set at 80%, the sample size required to detect differences was 265 patients in each group calculated with the pass 11 software (NCSS LLC, Kaysville, UT). Taking into account a loss-to-follow-up rate of about 15%, 610 patients in total were recruited to meet the minimum sample size requirement.

Intention-to-treat analysis was used for all enrolled patients. Continuous variables were presented as mean ± SD if data are normally distributed; otherwise, data were presented as median (interquartile range, IQR) and data plot. Categorical values were presented as numbers (percentages). The differences between the 2 groups were compared using the Mann-Whitney or χ^2 tests. The variables of comparison between POI and non-POI with significant differences ($P < .10$) in univariate analysis were included in multivariate analysis, which was used to identify the associated risk factors of POI occurrence. To further illustrate the effect of TEAS on POI, the variables that were significantly different between the TEAS group and sham group were used as the adjusting factors in multivariate analysis. The effectiveness of TEAS on POI was then stratified by other independent factors. The Kaplan-Meier log-rank test was used to illustrate the cumulated no flatus rate and the POI incidence. The center effect of POI incidence was analyzed by the Breslow-Day test. All statistical analyses were performed using the SPSS software (version 25.0, IBM Corporation, NY). All authors had access to the study data and reviewed and approved the final manuscript.

Results

Characteristics of study population

A total of 992 patients were scheduled for colorectal surgery at 10 hospitals in China between May 2018 and August 2019. Recruitment was terminated on August 31, 2019. During the 2-step screenings, 238 patients were excluded at the initial screening (including 142 patients who did not meet the study criteria and 96 patients who declined to participate), and 144 patients were excluded at the second screening (including 109 patients with intraoperative enterostomy, 18 patients whose surgeries were canceled, 4 patients who withdrew from the study, and 13 patients with other reasons). A total of 610 patients were enrolled and randomly divided into 2 groups: 307 patients allocated to the sham group and 303 patients to the TEAS group. All 610 patients were included in the intention-to-treat analysis (Fig 1). No TEAS-related adverse events (eg, skin allergy) occurred in the study. Patients had good TEAS treatment compliance with stimulating intensity in the PACU ranging from 13.0 mA to 15.0 mA, which was 2 to 5 mA stronger than that in the ward. Some patients in the TEAS group experienced muscle twitch (Supplementary Table S1 in Appendix 2).

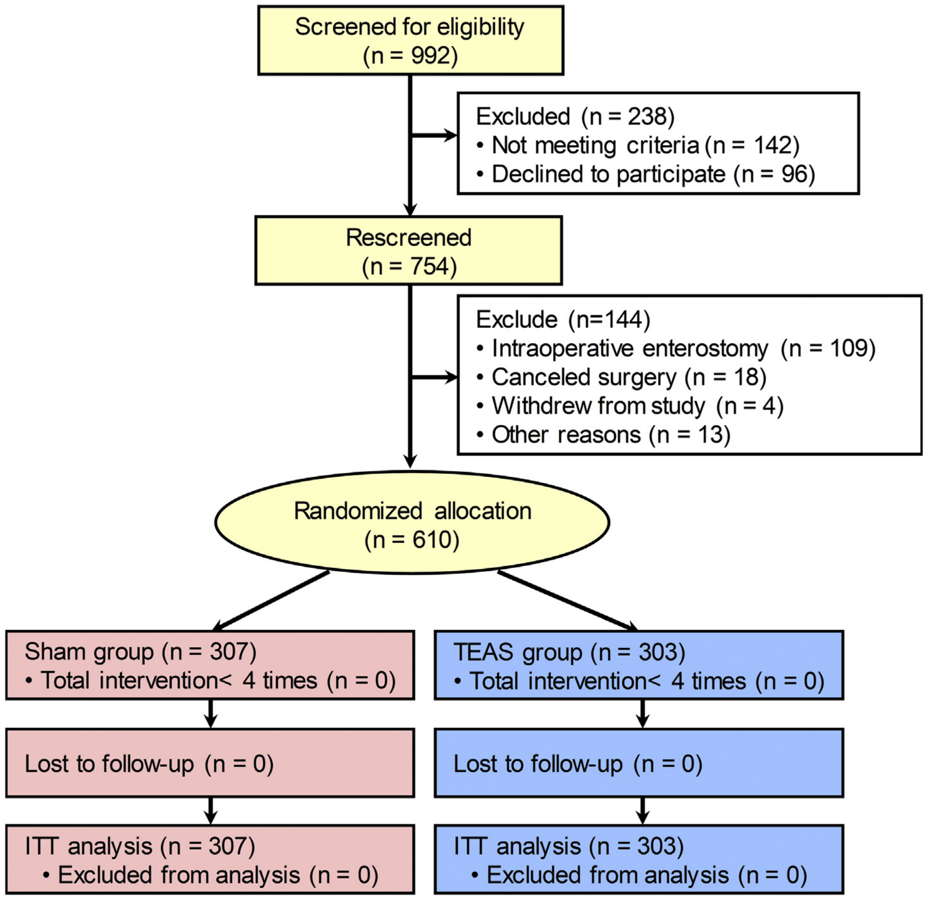

Fig 1. CONSORT diagram for patients enrolled in the study.
TEAS, transcutaneous electrical acupoint stimulation; *ITT*, intention-to-treat.

The demographics, comorbidities, surgical data, intraoperative loading volume, and perioperative anesthetic use between the TEAS and sham groups are presented in Tables I and II, and comparison of these variables between the POI and non-POI groups are presented in Supplementary Tables S2 and S3 in Appendix 2. The incidence of intraoperative blood plasma use in the TEAS group was significantly lower than that in the sham group, but there was no significant difference between the POI and non-POI groups. There was no statistically significant difference in the 30-day postoperative complications and the total length of hospital stay between the TEAS and sham groups (Supplementary Table S4).

Primary and secondary outcomes

Compared to the sham group, patients in the TEAS group had a significantly lower POI incidence (32.3% vs 41.0%, $P = .026$; OR, 0.68, 95% CI 0.48–0.96; $P = .029$) and earlier gastrointestinal function recovery (Fig 2). Patients in the TEAS group had a shorter time to first flatus by 12.8 hours (95% CI 7.5–18.2; $P < .001$), and the cumulative no flatus rate of the TEAS group was more significantly decreased compared to the sham group ($P = .002$). Compared to the sham group, patients in the TEAS group had first defecation earlier by 14.9 hours (4.4, 25.3; $P = .005$), first resumed normal diet earlier by 51.6 hours (14.8, 88.4; $P = .006$), and first bowel sounds heard earlier by 4.6 hours (0.9, 8.3; $P = .016$) (Table III). Average abdominal pain scores, distention scores, nausea scores, and incidence of vomiting at the postoperative 72 hours were not significantly different between the 2 groups. The *P* value of POI incidence was .873 in the Breslow-Day test, indicating the center homogeneity of POI incidence.

Multivariate analysis for POI

The univariate factors (Supplementary Tables S2 and S3 in Appendix 2) were analyzed to assess the POI risk. Only intraoperative blood plasma infusion was significantly different between the TEAS and sham groups (Tables I and II), and hence was used as an adjusting factor in multivariate analysis. Compared to the sham group, the TEAS group reduced the risk of POI occurrence (OR, 0.68, 95% CI 0.48–0.96; $P = .029$) by multivariate-unadjusted analysis; this risk reduction (OR, 0.69, 95% CI 0.49–0.98; $P = .038$) was still significant after adjusting for intraoperative plasma infusion usage (Supplementary Table S5 in Appendix 2). In multivariate-unadjusted analysis, patients who underwent laparoscopic surgery showed a reduction in POI risk (OR, 0.62, 95% CI 0.44–0.89; $P = .009$). Patients who underwent colectomy (OR, 2.01; 95% CI, 1.39–2.90; $P < .001$) and had a history of hypertension (OR, 1.54; 95% CI, 1.04–2.30; $P = .033$) had an increased POI risk. The delay in out-of-bed activities (OR, 1.02; 95% CI 1.01–1.02; $P = .001$) was associated with a higher risk of POI. Meanwhile, the TEAS group showed a statistically significant reduction in POI-associated risks without interaction with other independent factors, after respectively stratified with a history of hypertension, laparoscopic surgery, colectomy, and the time of first out-of-bed activities (all $P > .05$) (Supplementary Table S6 in Appendix 2).

Table I
Baseline characteristics

	TEAS (n = 303)	Sham (n = 307)
Age, years	62.0 (53.0–69.0)	63.0 (54.0–70.0)
Men	176 (58.1)	156 (50.8)
Body mass index, kg/m^2	22.5 (20.4–24.7)	22.7 (20.7–24.4)
Education level		
≤Primary education	74 (28.6)	84 (35.4)
Secondary education	146 (56.4)	122 (51.5)
Tertiary education	39 (15.1)	31 (13.1)
ASA score		
I	24 (7.9)	22 (7.2)
II	243 (80.2)	240 (78.2)
III	36 (11.9)	45 (14.7)
Comorbidities		
Coronary heart disease	17 (5.6)	20 (6.5)
Chronic obstructive pulmonary disease	18 (5.9)	18 (5.9)
Diabetes	31 (10.2)	30 (9.8)
Hypertension	80 (26.4)	63 (20.5)
Preoperative hypokalemia	252 (92.3)	254 (90.1)
Abdominal surgery history	17 (5.6)	21 (6.8)
Depth of infiltration		
T0	8 (2.8)	11 (3.7)
Tis	9 (3.2)	8 (2.7)
T1	12 (4.2)	11 (3.7)
T2	43 (15.1)	46 (15.5)
T3	107 (37.5)	112 (37.7)
T4	106 (37.2)	109 (36.7)
Lymph node metastasis		
N_0	182 (64.8)	185 (63.8)
N_1	99 (35.2)	105 (36.2)
Distant metastasis	13 (4.6)	13 (4.5)
Dukes' staging		
A	61 (22.1)	61 (21.3)
B	112 (40.6)	113 (39.5)
C	88 (31.9)	98 (34.3)
D	15 (5.4)	14 (4.9)

Data are presented as median (IQR) or patient's number (%).
ASA, American Society of Anesthesiologists physical status classification system; *IQR*, interquartile range; *TEAS*, transcutaneous electrical acupoint stimulation.

Table II
Surgery and perioperative anesthetics records

	TEAS (n = 303)	Sham (n = 307)	P value
Tumour size, cm	5.0 (3.0–6.0)	5.0 (4.0–6.0)	.873
Operation time, h	3.3 (2.6–4.1)	3.2 (2.5–4.2)	.687
Colectomy	190 (62.7)	179 (58.3)	.266
Right hemicolectomy	78 (25.7)	77 (25.1)	.851
Transverse colectomy	23 (7.6)	15 (4.9)	.167
Left hemicolectomy	22 (7.3)	20 (6.5)	.716
Sigmoidectomy	67 (22.1)	67 (21.8)	.932
Proctectomy	113(37.3)	128 (41.7)	.266
Laparoscopic	199 (65.7)	187 (60.9)	.222
Intraoperative blood loss, ml	100.0 (50.0–200.0)	100.0 (70.0–200.0)	.240
Intraoperative crystalloid fluid, L	1.7 (1.5–2.0)	1.5 (1.5–2.0)	.153
Intraoperative colloidal fluid, L	0.5 (0.5–1.0)	0.5 (0.5–1.0)	.178
Intraoperative red blood cell	30 (9.9)	31 (10.1)	.946
Intraoperative plasma	17 (5.6)	34 (11.1)	.015
Propofol, mg	800.0 (530.0–1050.0)	790.0 (500.0–1000.0)	.171
Etomidate, mg	14.0 (10.0–16.0)	14.0 (10.0–16.0)	.721
Sevoflurane, ml	30.0 (15.8–40.0)	30.0 (19.0–40.0)	.593
Dexmedetomidine, μg	90.0 (57.6–122.2)	90.0 (59.4–127.2)	.764
Remifentanil, μg	1500.0 (1071.0–2000.0)	1500.0 (1140.0–2000.0)	.744
Sufentanil, μg	30.0 (25.0–35.0)	30.0 (25.0–30.0)	.643
Cisatracurium bromide, mg	22.0 (15.0–30.0)	23.0 (16.0–31.0)	.558
Rocuronium bromide, mg	50.0 (0.0–60.0)	53.0 (0.0–60.0)	.871
Postoperative analgesia	297 (98.0)	304 (99.0)	.304
Delayed out-of-bed activities, hr	34.5 (20.9–43.8)	38.2 (23.8–44.7)	.018

Data are presented as median (IQR) or number of patient (%).
IQR, interquartile range; *TEAS*, transcutaneous electrical acupoint stimulation.

Serum biomarkers

The TEAS group had higher concentrations of Ach (median [IQR]: 124.5 [80.4, 171.7] pg/mL vs 80.4 [64.7, 124.5] pg/mL, $P = .007$) and IL-10 (157.0 [97.4, 241.3] pg/mL vs 83.6 [49.0, 127.3] pg/mL vs, $P < .001$) but had lower concentrations of IL-6 (15.8 [11.6, 25.5] pg/mL vs 30.0 [19.8, 44.0] pg/mL, $P < .001$) and iNOS (7.2 [3.4, 12.1] IU/mL vs 12.3 [7.7, 17.4] IU/mL, $P = .002$) than did

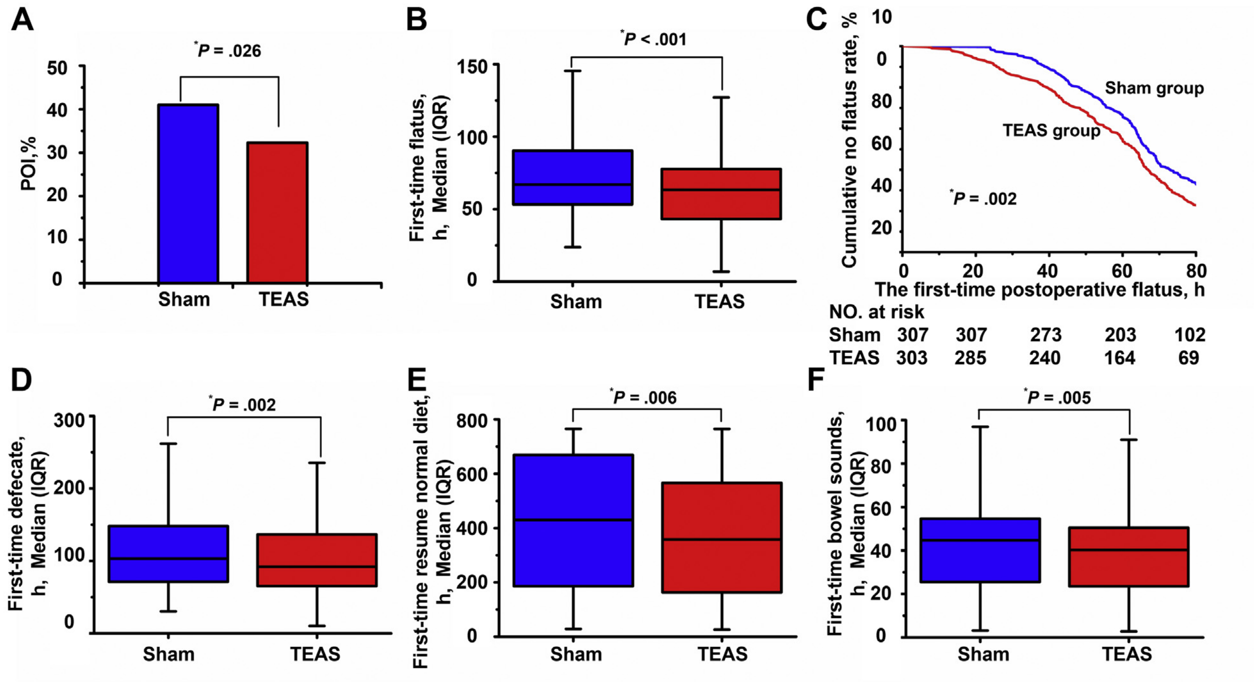

Fig 2. Postoperative paralytic ileus (POI) and overall gastrointestinal functional recovery. (A) POI incidence, (B) first-time flatus, (C) Kaplan-Meier curves for cumulative no flatus rate, (D) first-time defecation, (E) first-time resume normal diet, and (F) the first-time bowel sounds. The data are presented as number of patients (%), mean (SD), and cumulative curves ($n = 303$ in the TEAS group; $n = 307$ in the sham group); $^*P < .05$ versus the sham group.

Table III
Primary and secondary outcomes

Outcomes	TEAS (n = 303)	Sham (n = 307)	Mean difference (95% CI)	P value
Primary outcome				
POI, *n* (%)	98 (32.3)	126 (41.0)		.026
Secondary outcomes (h)				
First-time flatus	63.4 ± 29.0	76.3 ± 37.9	−12.8 (−18.2 to −7.5)	.001
First-time defecation	106.7 ± 65.6	121.5 ± 65.6	−14.9 (−25.3 to −4.4)	.005
First-time resume normal diet	373.9 ± 224.9	425.5 ± 235.6	-51.6 (−88.4 to −14.8)	.006
First-time bowel sounds	41.4 ± 22.6	46.0 ± 24.1	−4.6 (−8.3 to −0.9)	.016
Abdominal pain score	1.4 ± 1.4	1.3 ± 1.5	0.1 (−0.2 to 0.3)	.555
Distention score	1.6 ± 0.9	1.6 ± 0.9	−0.1 (−0.2 to 0.1)	.389
Nausea score	1.0 ± 1.6	1.1 ± 1.7	−0.1 (−0.3 to 0.2)	.624
Number of vomiting episodes	0.0 ± 0.5	0.0 ± 0.1	0.0 (−0.0 to 0.1)	.155

Data are presented as mean ± SD or number of patients (%).
POI, postoperative paralytic ileus.

the sham group (Fig 3; $n = 31$/group). Similarly, the concentrations of Ach and IL-10 in the POI group were lower than those in the non-POI group, whereas the concentrations of IL-6 and iNOS in the POI group were higher than those in the non-POI group (Supplementary Table S7 in Appendix 2). All the biomarkers measured at the postoperative 72 hours between laparoscopic surgery and proctectomy surgery were not statistically significant (Supplementary Table S8 in Appendix 2).

Discussion

POI is a transient cessation of coordinate bowel motility after surgical intervention. After nonalimentary surgery, the gastrointestinal functional recovery usually occurs within 72 hours.[25] The enhanced recovery of intestinal motility after colorectal surgery was one of the elements documented in several Enhanced Recovery After Surgery (ERAS) protocols.[26] The comprehensive ERAS protocols after surgery lowered the POI incidence by 4%,[27] and Alvimopan reduced POI incidence by 3.8%.[28] Although the positive effect of TEAS or electroacupuncture on POI has already been suggested,[19,29] it still has not been adequately verified due to an unclear POI definition, underpowered small sample sizes, and unclear underlying mechanism.[20] Furthermore, most studies have only reported the gastrointestinal function recovery, such as first flatus, defecation, or other continuous variables.[19,29] Only 3 trials reporting the POI incidence were included until January 2017 from 6 databases, and the highest reduction rate was only 4.2%.[24] Therefore, we performed this large-sample-size, multicenter, evaluator-blinded, randomized controlled study to investigate the effects of TEAS on the incidence of POI after colorectal surgery[21] and the possible underlying mechanisms. We found that the POI incidence was significantly reduced by 8.7% with TEAS treatment. TEAS also decreased the risk of POI occurrence by 32% (OR, 0.68, 95% CI 0.48–0.96; $P = .029$) and shortened the time to first flatus by 12.8 hours ($P < .001$). Furthermore, TEAS treatment significantly shortened the defecation recovery, resumption of bowel sounds, and normal diet times. TEAS also significantly increased plasma Ach and IL-10, but it decreased IL-6 and iNOS, indicating that this

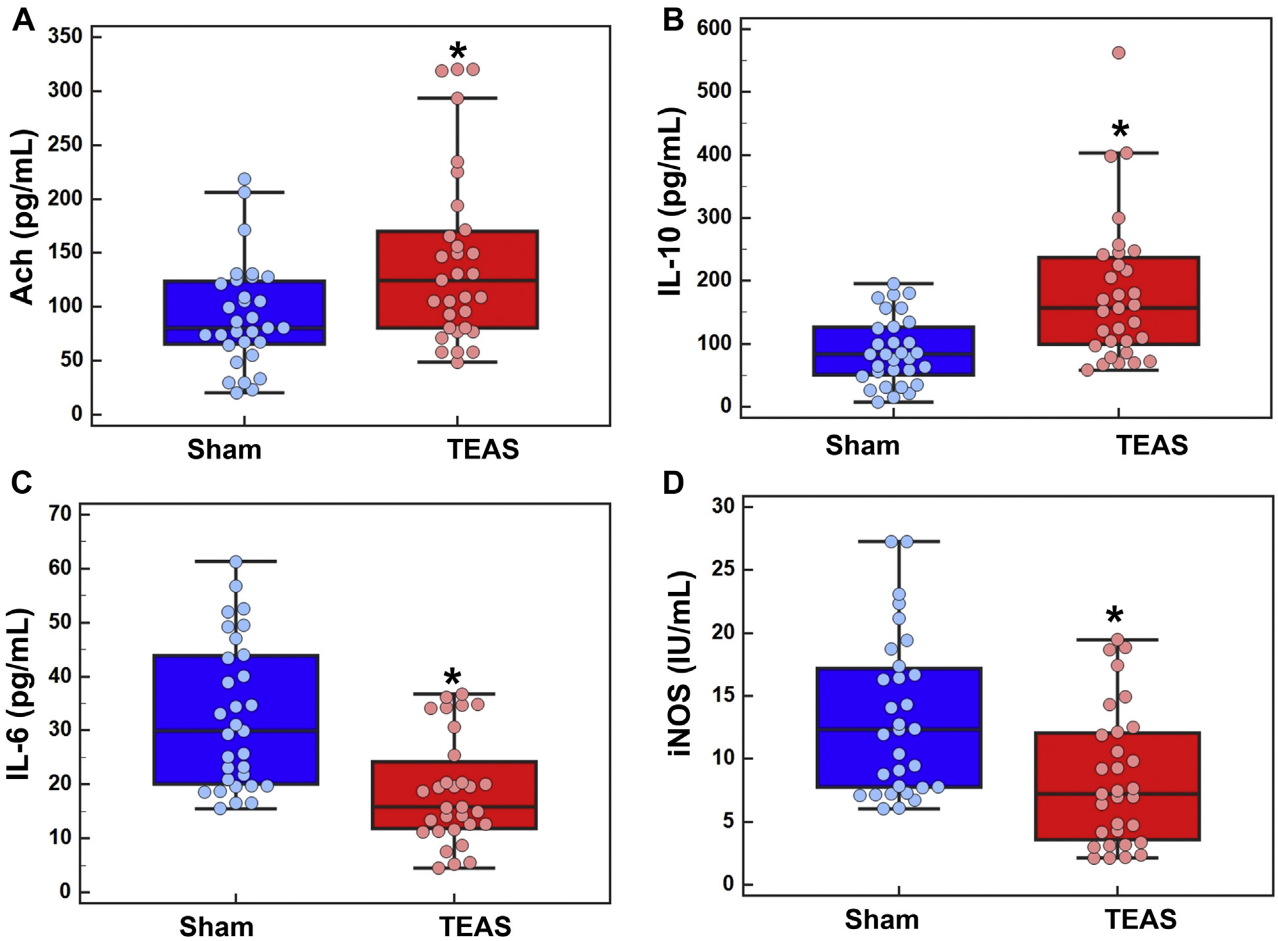

Fig 3. The biomarkers measured at postoperative 72 hours. (A) Acetylcholine (Ach); (B) interleukin-10 (IL-10); (C) interleukin-6 (IL-6); (D) inducible nitric oxide synthase (iNOS); the data are presented as median (IQR) and dot plot (n = 31/group). $^*P < .05$ versus the sham group.

intervention increased parasympathetic nerve tone and anti-inflammatory actions to prevent POI.

The noxious stimuli of surgery activate the sympathetic but inhibit the parasympathetic system. Surgery also induces local and systemic inflammation[30]—specifically, the resident and infiltrating macrophages release proinflammatory cytokines, which, in turn, recruit neutrophils to infiltrate the intestinal wall muscle.[31] As a result, the autonomic nervous system imbalance, inflammation, and beyond result in POI after surgery. Our data showed that TEAS reduced the risk of POI by 32%. This was further demonstrated in the Kaplan-Meier curve, the log-rank test, and adjusted and stratified analyses (Fig 2; Supplementary Tables S5 and S6 in Appendix 2). The underlying mechanisms are yet to be explored, but previous studies have demonstrated that acupuncture can increase gastric acid secretion through "nerve reflex" of the somatic nerves of the afferent pathway and the vagal branch innervates the stomach as the efferent pathway.[12] TEAS likely modulates the imbalance between the sympathetic and parasympathetic system to reduce POI via the neurofunctional changes of nerves nuclei.[32] Indeed, TEAS did increase Ach in the plasma measure at 72 hours after surgery in the patients in our study, indicating that, as an intestinal excitatory neurotransmitter, Ach was released due to the vagal nerve activation induced by acupuncture. iNOS is the main source of NO in inflammatory response,[33] which disrupts the pacemaker activity of interstitial Cajal cells and contributes to POI development.[34] The decreases in IL-6 and iNOS reported in our study might have been induced by the cholinergic anti-inflammatory pathway through the TEAS triggered vagal nerve activation via alpha-7 nicotinic receptors to downregulate the inflammatory cytokine production.[35] Unlike IL-6 and iNOS, IL-10 is an anti-inflammatory cytokine[8] that was increased in our patients treated with TEAS. This suggests that the decreased incidence of POI seen with TEAS treatment might be due to increasing vagal nerve tone and its anti-inflammatory response (Fig 4).

The POI incidence in our sham group was 41%, which was higher than previously reported by up to 15%.[36,37] These differences may be due to the different diagnostic criteria being used. POI is characterized by various symptoms including delayed flatus and defecation, abdominal pain, bloating, nausea, vomiting, and intolerance to an oral diet. The most commonly used criteria were the passage of flatus (83%) in the 47 publications previously reported.[6] Our trial was, therefore, set with no flatus >72 hours as a criterion, whereas other studies only included the moderate-to-severe symptoms of POI.[27] Based on the results from our trial, the following could be concluded: (1) The acupoints ST36, ST37, and SP6 (Supplementary Fig S1 in Appendix 2) rather than a single acupoint were used. ST36-SP6 was reported to significantly shorten the postoperative flatus and defecation time, whereas ST36-ST37 reduced inflammatory reaction and improved intestinal function.[38] (2) The first TEAS intervention was performed in the PACU because early active intervention might effectively inhibit the inflammatory cascade within the postoperative 6 hours.[8] (3) Given that morning time is the highest period of vitality for gastrointestinal motility in circadian rhythm,[39] the TEAS treatment was carried out from 9 AM to 11

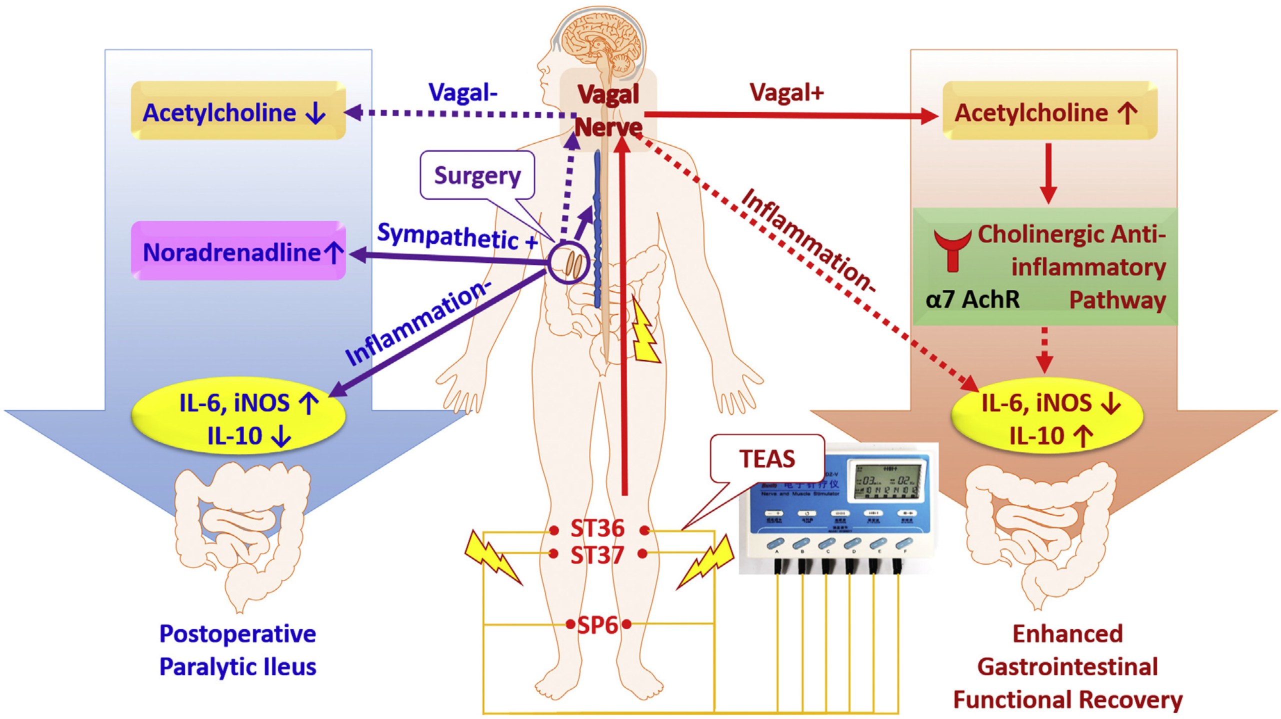

Fig 4. The illustration of underlying mechanisms of transcutaneous electrical acupoint stimulation (TEAS) applied in lower limbs in reducing the POI incidence. Surgical trauma and noxious stimuli induced POI through vagal nerve inhibition, sympathetic activation, and inflammation, whereas TEAS enhanced gastrointestinal functional recovery via vagal nerve activation and inflammation inhibition, showing as increased plasma Ach and IL-10 and decreased IL-6 and iNOS.

AM on the first day to third day after surgery. (4) The TEAS intervention was individually adjusted to the maximum tolerance intensity for half an hour at each treatment, which was consistent with the current standard of acupuncture treatment.[24]

In our study, we found that 5 independent perioperative factors of TEAS (preoperative hypertension, laparoscopic surgical technique, colectomy surgery, and postoperative out-of-bed activity) were related to POI (Supplementary Table S5 in Appendix 2). The POI incidence in patients with hypertension was 1.5 times higher than those without hypertension, because hypertension increases the permeability of the gut epithelial barrier and inflammatory response.[40] Regarding surgical factors, our study demonstrated that laparoscopic surgery reduced the risk of POI by 38% compared to open surgery. Similar results have been reported that the first bowel movement was 1 day earlier after laparoscopic colectomy than after open colectomy.[41,42] This finding is likely due to the laparoscopic technique reducing the postoperative atony of the small bowel, attenuating stress responses, and improving active motility at the splenic flexure of the colon. Furthermore, our data also showed that colectomy has a 2-fold higher risk of POI than rectal surgery, which is consistent with the incidence of bowel dysfunction in colectomy (18.7%) and proctectomy (8.3%) in another prospective trial.[43] This might be because the colon plays a more important role in gastrointestinal motility than the rectum. In line with multicenter observational studies and meta-analyses on ERAS protocols of early mobilization,[44] our data also demonstrated that delayed out-of-bed activities increased the risk of POI.

The strengths of our trial are a relatively large sample size, with patients recruited from multiple centers, and the complementary biomarkers being measured in supporting the preventive effectiveness of TEAS on the development of POI. However, this trial is not without limitations. First, although patients were blinded to the group allocation, patients could not be blinded to the treatment due to the nature of the intervention, because they can sense the acupoint stimuli in the TEAS group compared to no treatment in the sham group. However, the psychotherapeutic effects of sham acupuncture can be effective.[45] If that is the case in our study, then the reduced incidence of POI by TEAS therapy would be underestimated, and hence, the efficacy of TEAS in treating POI can be assured. Second, all our patients underwent colorectal surgery. Whether TEAS is also effective in decreasing the incidence of POI in other kinds of surgery, such as upper abdominal surgery, is unknown. Third, the study excluded patients with prophylactic ileostomy due to the difficulties in the evaluation of flatus and the effects of TEAS for these patients were unknown. Fourth, block randomization in a 1:1 ratio with a block size of 4 could ensure that the number of patients in the 2 groups were essentially equal with highly efficient enrollment; however, the small block size may not completely avoid the violation of allocation concealment. Finally, this study was not undertaken in combination with the comprehensive ERAS program, which is worth pursuing in future studies. Nevertheless, the data derived from our multicenter clinical trial clearly demonstrated that TEAS decreased the incidence of POI and promoted postoperative gastrointestinal functional recovery after surgery; hence, the technique should be considered for use in clinical practice. In line with our data, nonpharmacological or physiotherapies such as electric nerve stimulation or TEAS used in our study have been well documented to provide enormous therapeutic benefits for functional gastrointestinal disorders in other disease types.[46,47]

In conclusion, this multicenter randomized controlled study demonstrated that TEAS significantly lowered POI incidence (no flatus for >72 hours) and earlier gastrointestinal function recovery (the first-time indicator appeared in advance), which may be due to its increasing parasympathetic nerve tone and anti-inflammatory actions. The clinical significance of TEAS in reducing the incidence of POI is important because a rapid gut function recovery after colorectal surgery is beneficial. However, whether TEAS treatment improved long-term surgical outcomes remains unknown and warrants further study.

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Conflict of interest/Disclosure

The authors have no competing interests.

Supplementary materials

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